

Multiple Choice

1. **Answer:** D

Options: A) Gustatory receptors; B) Photoreceptors; C) Auditory receptors; D) Proprioceptors; E) Thermoreceptors

Note: Proprioceptors provide the brain with information about body position and movement, allowing us to perform manual tasks without visual input.

2. **Answer:** A

Options: A) thrombin; B) glycosaminoglycans; C) collagens; D) fibroblasts

Note: Thrombin is a blood coagulation factor involved in the clotting process and is not a component of connective tissue, which primarily consists of cells, fibers, and ground substance.

3. **Answer:** B

Options: A) Vinegar has a lower concentration of hydrogen ions than hydroxide ions; B) Vinegar has an excess of hydrogen ions with respect to hydroxide ions; C) Vinegar has a higher concentration of hydroxide ions than water; D) Vinegar has no hydroxide ions; E) Vinegar does not contain hydrogen ions

Note: Vinegar is classified as an acid because it has a higher concentration of hydrogen ions (H) compared to hydroxide ions (OH), indicating an excess of H which defines acidic solutions.

4. **Answer:** A

Options: A) Protects the cell from foreign invaders; B) Acts as a storage for genetic material; C) Regulates the pH level within the cell; D) Transports oxygen throughout the cell; E) Removes nitrogenous wastes

Note: The contractile vacuole primarily regulates osmotic pressure and expels excess water from the cell, which indirectly protects the cell from potential damage caused by foreign invaders that may exploit osmotic imbalances.

5. **Answer:** A

Options: A) their rate of mutation should be identical.; B) they should have the exact same DNA structure.; C) they should have identical physical characteristics.; D) they should share the same number of chromosomes.; E) they live in very different habitats.

Note: The correct answer is based on the understanding that while distantly related organisms may have evolved different traits and adaptations, the fundamental processes of DNA replication and mutation rates are generally conserved across species, leading to similar mutation rates regardless of evolutionary distance.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The analyses show that structural characteristics of a practice are not associated with uptake of a new IT facility, but that its use may be influenced by post-graduate education in the relevant clinical condition. For this diabetes system at least, practice nurse use was critical in spreading uptake beyond initial GP enthusiasts and for sustained and rising use in subsequent years.

7. **Answer:** True

Options: A) True; B) False

Note: Our data suggest that the SCL 90-R is best viewed as an indicator of unidimensional emotional distress and somatic effects of structural brain injury.

8. **Answer:** True

Options: A) True; B) False

Note: Regarding the pedigree, we discuss different modes of inheritance. The presence of consanguineous unions in this family suggests the possibility of a common ancestor and thus a recessive autosomal mode of inheritance.

9. **Answer:** False

Options: A) True; B) False

Note: According to our study, there is a great variability when different brands and models of scanners are compared directly. Furthermore, the CT scan analysis and HU evaluation appears to gather insufficient information in order to characterize and identify the composition of renal stones.

10. **Answer:** True

Options: A) True; B) False

Note: Self-reported mechanical factors associated with chronic oro-facial pain are confounded, in part, by psychological factors and are equally common across other frequently unexplained syndromes. They may represent another feature of somatisation. Therefore the use of extensive invasive therapy such as occlusal adjustments and surgery to change mechanical factors may not be justified in many cases.

Long Form Response

11. **Answer:** Molecular cloning presents several potential hazards that have significant implications for public health, biosecurity, and ethical considerations in genetic research. One major concern is the risk of creating virulent strains of *E. coli* or other pathogens that could lead to outbreaks of infectious diseases, posing a threat to community health. Additionally, the potential military applications of genetically modified bacteria raise biosecurity issues, as engineered organisms could be weaponized for harmful purposes. Furthermore, the introduction of cancer-causing genes into human cells through cloning techniques raises profound ethical questions about the long-term impacts on human

health and the potential for unintended consequences in genetic manipulation. These factors underscore the necessity for stringent regulations and ethical guidelines in the field of molecular cloning to mitigate associated risks.

Note: correct because it identifies specific hazards of molecular cloning, such as the creation of virulent pathogens and the potential for biosecurity threats, which are critical concerns for public health and ethical standards in genetic research.

12. **Answer:** Cyclic electron flow through photosystem I (PSI) plays a crucial role in the photosynthetic process of eukaryotic cells by generating ATP without the production of NADPH. During this process, electrons are excited by light energy absorbed by PSI and are subsequently transferred through a series of carrier proteins in the thylakoid membrane, ultimately returning to PSI. This cyclical pathway allows for the establishment of a proton gradient, which drives ATP synthesis via ATP synthase. Although cyclic electron flow does not directly reduce NADP⁺ to NADPH, it complements the linear electron flow that does, by ensuring that ATP levels are sufficient for the Calvin cycle. The balance between these two pathways is essential for optimizing energy production and the overall efficiency of photosynthesis.

Note: correct because it accurately describes cyclic electron flow through photosystem I as a mechanism that generates ATP independently of NADPH production, emphasizing the importance of the proton gradient and ATP synthesis in the photosynthetic process.